

International Economic Development

ECON 4660/8666

Lecture 6 Education

Human Capital

- Human capital: the accumulated stock of skills and education that a worker has at a point in time.
 - Human capital refers to the intangible assets of an individual, distinct from physical assets such as machinery or buildings.
 - It is the collective sum of an individual's knowledge, skills, experiences, and abilities, which can be applied to produce economic value.
 - Unlike physical capital which depreciates over time (e.g., machinery wearing out), human capital can appreciate with experience and further education.

Human Capital

- It is an investment:
 - just like investing in stocks or real estate, investing in human capital (through education, training, or acquiring new skills) can yield returns in the form of higher income, better job opportunities, or improved job performance.
- The knowledge, the key input of human capital, is nonrivalry;
 - A good is considered "nonrivalrous" when its consumption by one individual does not reduce its availability to others. In the context of knowledge, this means that when one person learns or acquires knowledge, it doesn't diminish the ability of others to learn or access that same knowledge.
- Human capital investment has constant return to scale.

A simple model of human capital

Consumption is equal to total wage income.

$$C = wuH^s$$

Human Capital Accumulation

Future human capital depends on current human capital and time devoted to training and education.

$$H^{s'} = b(1 - u)H^s$$

Production Function

$$Y = zuH^d,$$

Firm's Profits

$$\pi = zuH^d - wuH^d = (z - w)uH^d.$$

Equilibrium Consumption

$$C = zuH$$

Human Capital Growth

Human capital evolves in equilibrium according to:

$$H' = b(1 - u)H$$

Equilibrium Human Capital Growth

Equilibrium growth rate of human capital:

$$\frac{H'}{H} - 1 = b(1 - u) - 1$$

Economic Policy and Growth

- Change b :
 - Make the education system more efficient.
 - Provide incentive into the school system.
- Change u :
 - Provide taxes or subsidies to education

I. Education

- Definition: All forms of human learning or narrowly, schooling.
 - Education encompasses a wide range of learning experiences, from formal settings like schools and universities to informal settings like learning from experiences, mentorships, or self-study.
- Some Facts
 - Demand for education is high
 - Education and income are positively related
 - Additionally, a more educated workforce can drive innovation and productivity, leading to economic growth and, consequently, higher wages.
 - Education expenditure is large in government budget. 5% GDP, world average

United States Federal, State and Local Government Spending

GDP: \$21,060.5

bln

GO:

\$36,562.0 bln

percent GDP

2020

FY 13 Fed Budg

full screen

state, local

United States

No Pie Chart

US CA >

Pop: 331.5 million

United States Federal

State and Local Government Spending

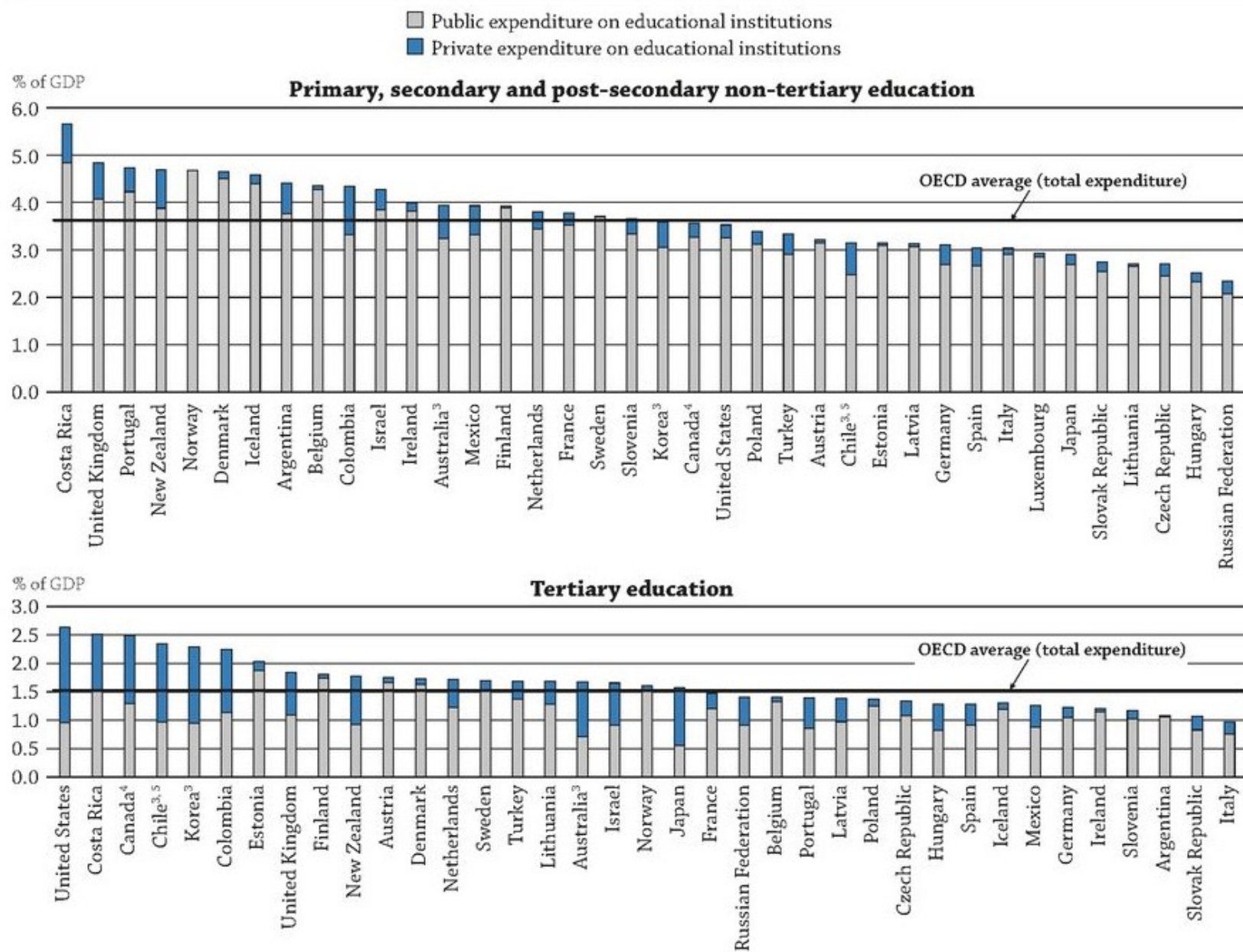
-5yr -1yr Fiscal Year 2020 in percent GDP +1yr +5yr

View: people *old* function radical census
programs altprog oldprog COFOG

	Fed	Gov. Xfer	State	Local	Total	chart
[+] Pensions	5.46	0.00	1.32	0.30	7.08	 
[+] Health Care	7.24	-2.32	3.83	0.91	9.65	 
[-] Education: <i>Start chart</i>	1.19	-0.29	1.67	3.77	6.34	 
[+] Pre primary thru secondary education	0.24	0.00	0.04	3.49	3.77	 
[+] Tertiary education	0.73	0.00	1.30	0.22	2.25	 
[+] Education not definable by level	0.22	-0.29	0.33	0.06	0.32	 
[+] Subsidiary services to education	0.00	0.00	0.00	0.00	0.00	 
[+] R and D Education	0.00	0.00	0.00	0.00	0.00	 
[+] Education n.e.c.	0.00	0.00	0.00	0.00	0.00	 
[+] Defense	4.80	0.00	0.01	0.00	4.81	 
[+] Welfare	5.27	-0.62	1.00	0.51	6.17	 
[+] Protection	0.34	-0.06	0.51	1.11	1.90	 
[+] Transportation	0.69	-0.33	0.72	0.87	1.95	 
[+] General Government	0.87	-0.02	0.20	0.31	1.35	 
[+] Other Spending	3.62	-0.26	0.46	1.99	5.82	 
[+] Interest	1.64	0.00	0.21	0.33	2.18	 
[+] Balance	0.00	0.00	0.04	0.00	0.04	 
[+] Total Spending	31.12	-3.90	9.96	10.11	47.29	 
[+] Federal Deficit	14.87	0.00	0.00	0.00	14.87	 
[+] Gross Public Debt	127.74	0.00	5.62	9.89	143.25	 

Figure B2.2. Public and private expenditure on educational institutions, as a percentage of GDP, by level of education (2013)

From public¹ and private² sources, by level of education and source of funds



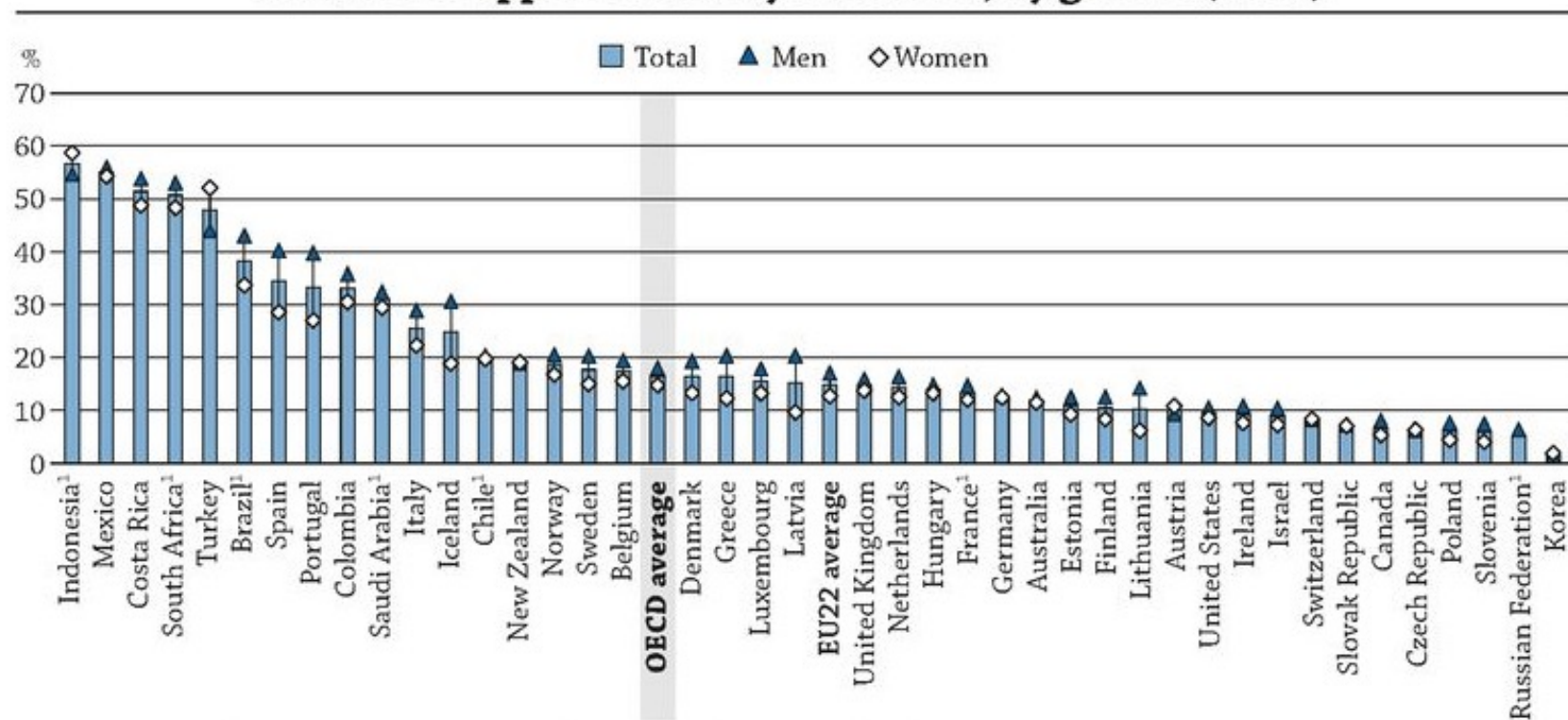
Public Spending on Educational Institutions

- Primary Education:
 - Typically, countries invest heavily in primary education, recognizing its foundational importance. It's not uncommon for countries, especially developing ones, to allocate a significant portion of their education budget to primary education.
 - In many countries, public expenditure on primary education can range from 1% to 3% of GDP.
- Secondary Education:
 - Secondary education also receives significant investment, though the exact percentage can vary widely based on the country's education system structure and priorities.
 - On average, countries might spend between 1% and 2.5% of their GDP on secondary education.
- Tertiary Education:
 - Public expenditure on tertiary education varies considerably across countries. In nations with publicly funded higher education systems, this percentage can be relatively high.
 - On average, public expenditure on tertiary education can range from 0.5% to 1.5% of GDP, though this is a broad estimate. as a percentage of GDP by level of education

Private Spending on Educational Institutions

- Primary Education:
 - Private expenditure on primary education is generally lower than secondary and tertiary levels, especially in countries where primary education is predominantly publicly funded.
 - It might range from 0.1% to 1% of GDP, with higher figures in countries with robust private primary education sectors.
- Secondary Education:
 - The private sector's role in secondary education varies by country. In some nations, private high schools or secondary institutions play a significant role.
 - Private expenditure might range from 0.5% to 2% of GDP.
- Tertiary Education:
 - Tertiary education often sees a more substantial private sector role, especially in countries with private universities and colleges.
 - Private expenditure on tertiary education can range widely, from 0.5% to 3% of GDP or even more, especially in countries where higher education is predominantly private.

Figure A1.1. Percentage of 25-34 year-old adults with below upper secondary education, by gender (2015)



Percentage w/ Below Upper Secondary Education

- Globally, there has been a trend of increasing educational attainment over the past few decades.
- The increase in educational attainment can be attributed to factors such as global initiatives promoting education, increased investment in education by governments, and societal recognition of the importance of education.
- By Gender:
 - Males: In many regions, especially in developed countries, the percentage of young males with below upper secondary education has been decreasing. However, there can be disparities based on regions and individual countries. For example, in some areas, male dropout rates in secondary education might be higher, leading to a larger percentage with below upper secondary education.
 - Females: Historically, in many parts of the world, females had lower educational attainment than males. However, there has been a significant push in recent decades to promote female education. As a result, in many countries, young females have caught up or even surpassed their male counterparts in terms of educational attainment. This trend means that in several regions, the percentage of young females with below upper secondary education has been decreasing at a faster rate than for males. However, disparities still exist in some areas due to cultural, societal, or economic factors.

Percentage w/ Below Upper Secondary Education

- Regional Variations:
 - The percentage of 25-34 year-olds with below upper secondary education can vary significantly by region.
 - Developed countries: In many developed nations, the majority of this age group has at least an upper secondary education, with relatively small percentages not achieving this level.
 - Developing countries: The picture can be more varied, with some nations seeing significant portions of this age group without upper secondary education, especially in regions where access to secondary education is still a challenge.

Figure A1.2. Percentage of adults scoring at literacy proficiency Level 4 or 5, by educational attainment (2012 or 2015)

Survey of Adult Skills, 25-64 year-old non-students

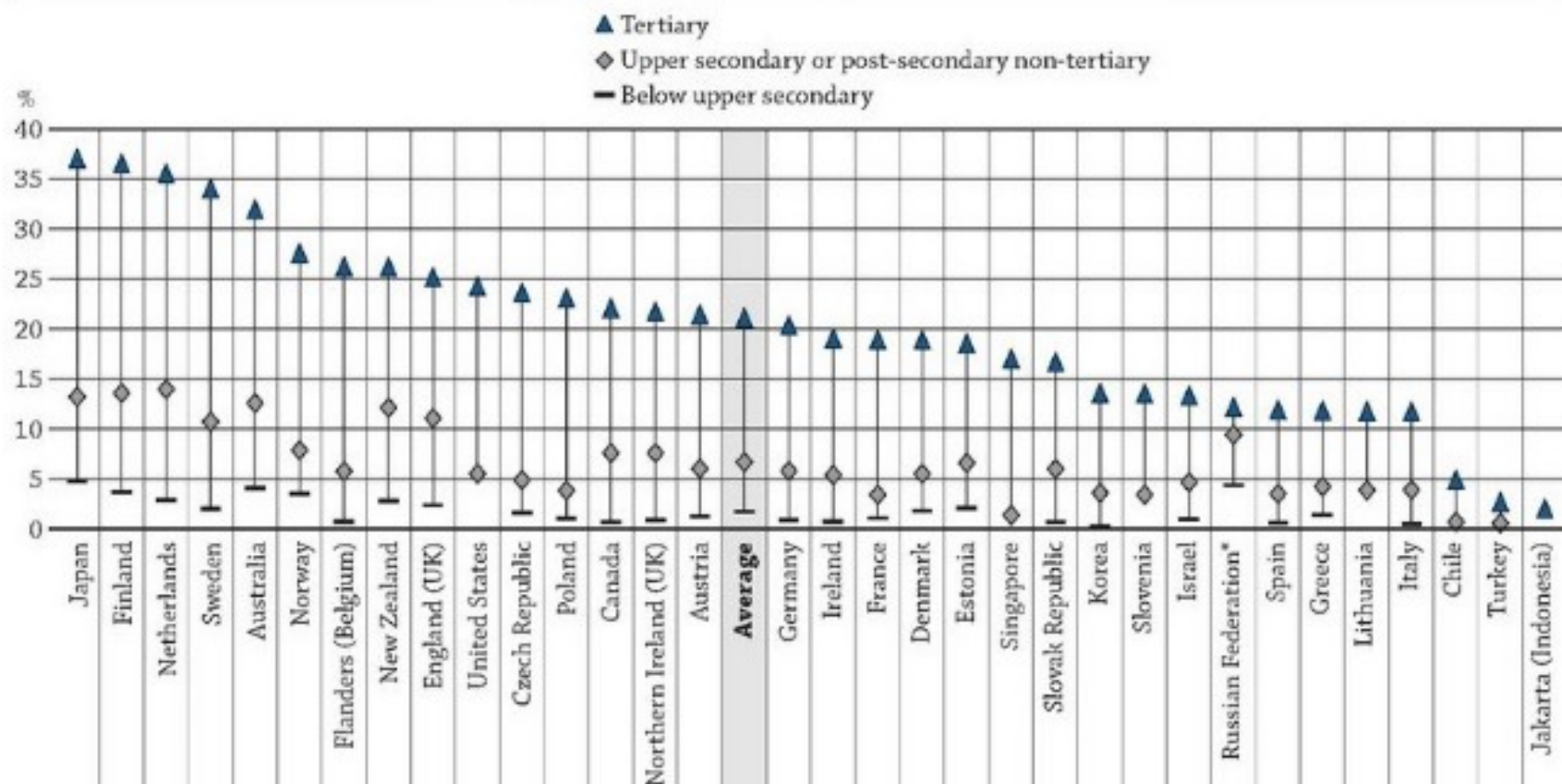


Figure A1.3. Percentage of 25-34 year-olds whose highest level of education is upper secondary or post-secondary non-tertiary, by programme orientation (2015)

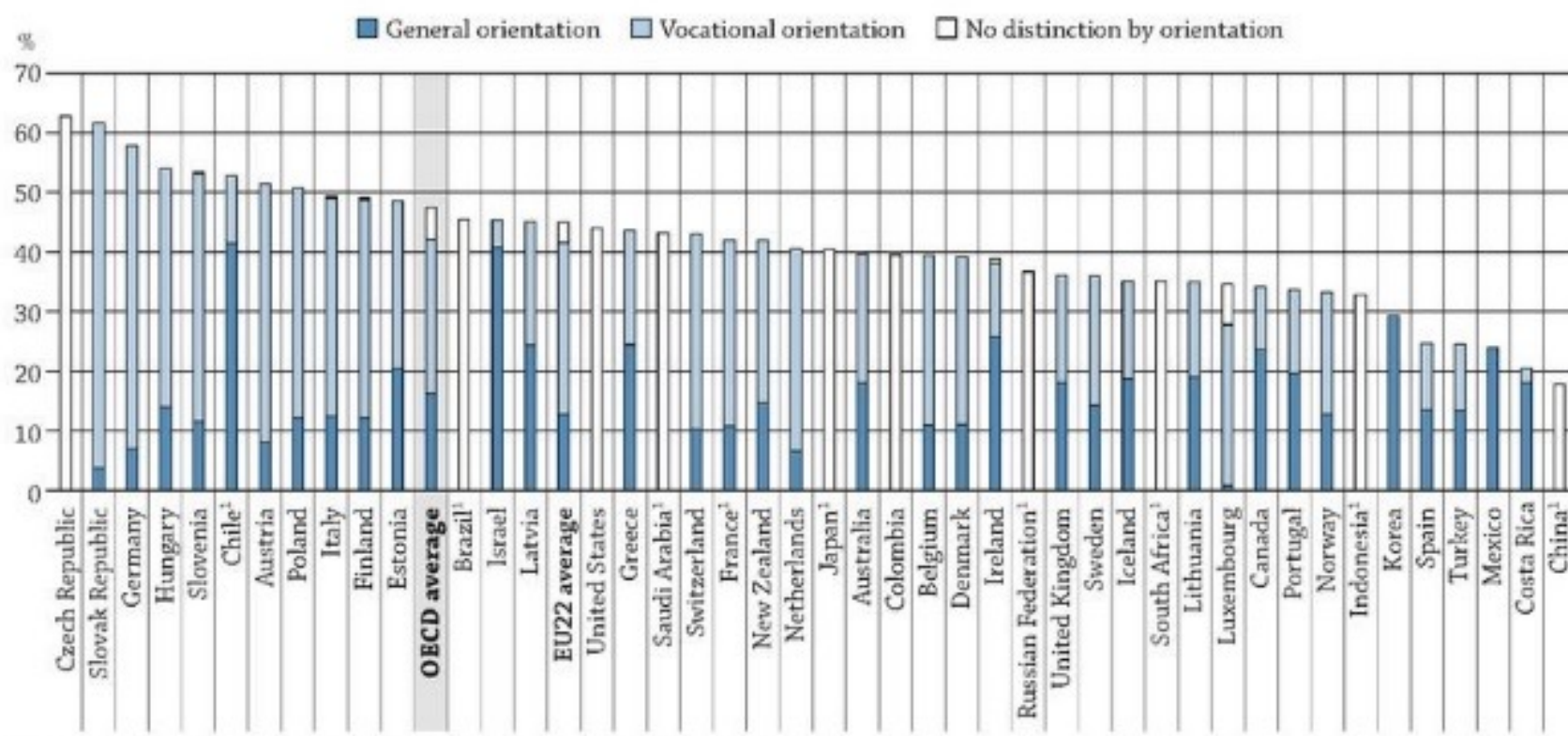


Figure A1.4. Percentage of 25-64 year-olds with tertiary education, by level of tertiary education (2015)

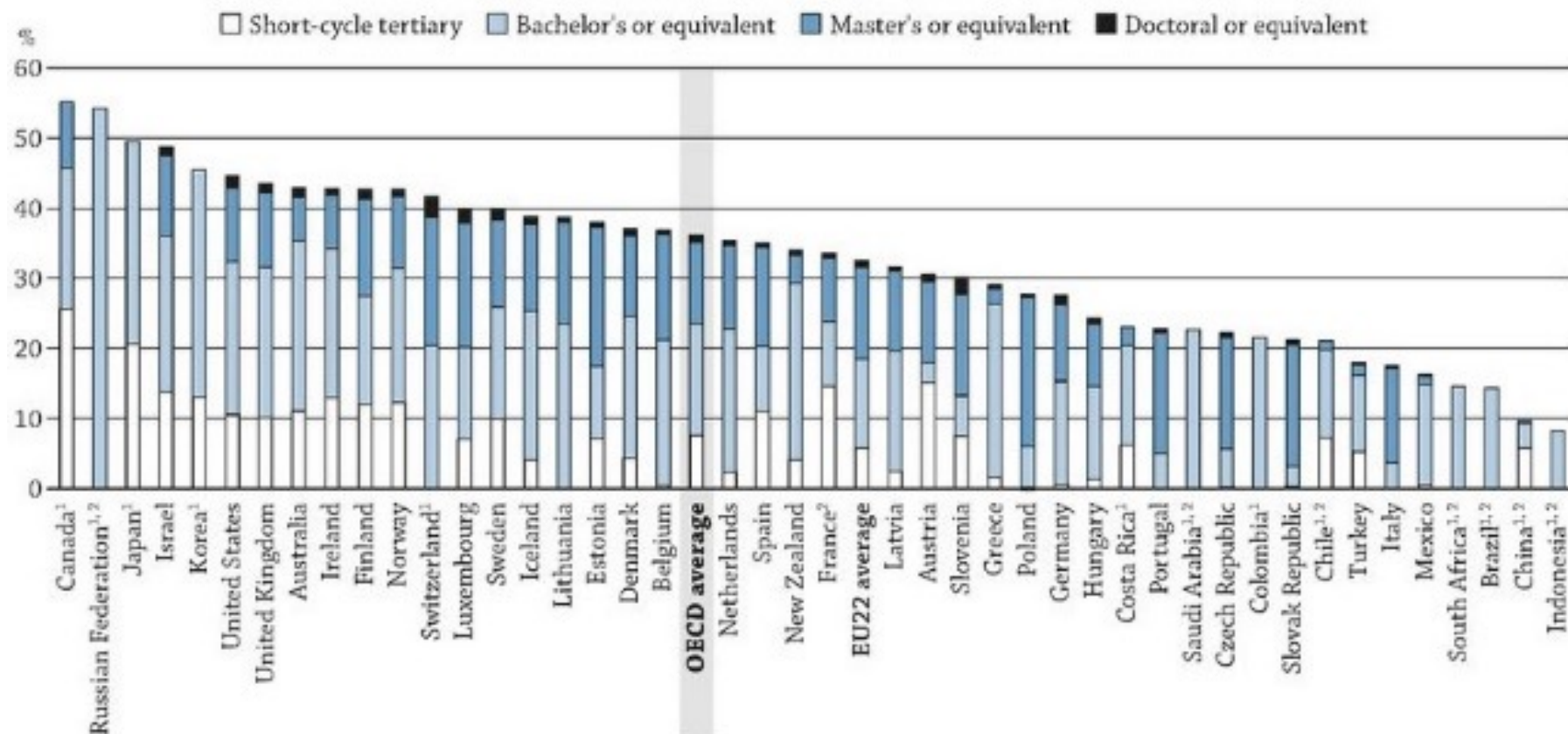
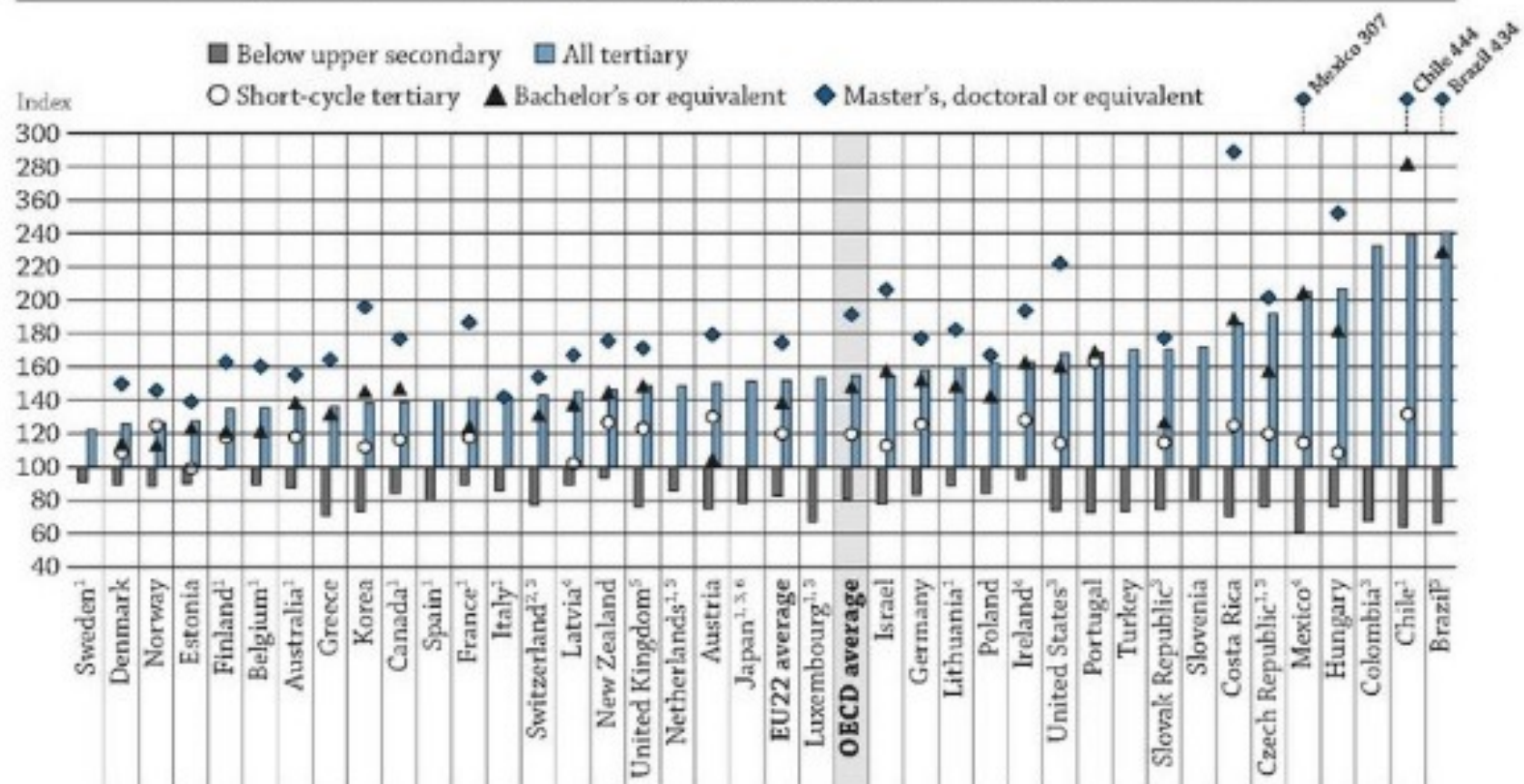


Figure A6.1. Relative earnings of adults working full time, by educational attainment (2014)

25-64 year-olds with income from employment; upper secondary education = 100



Relative Earnings by Educational Attainment:

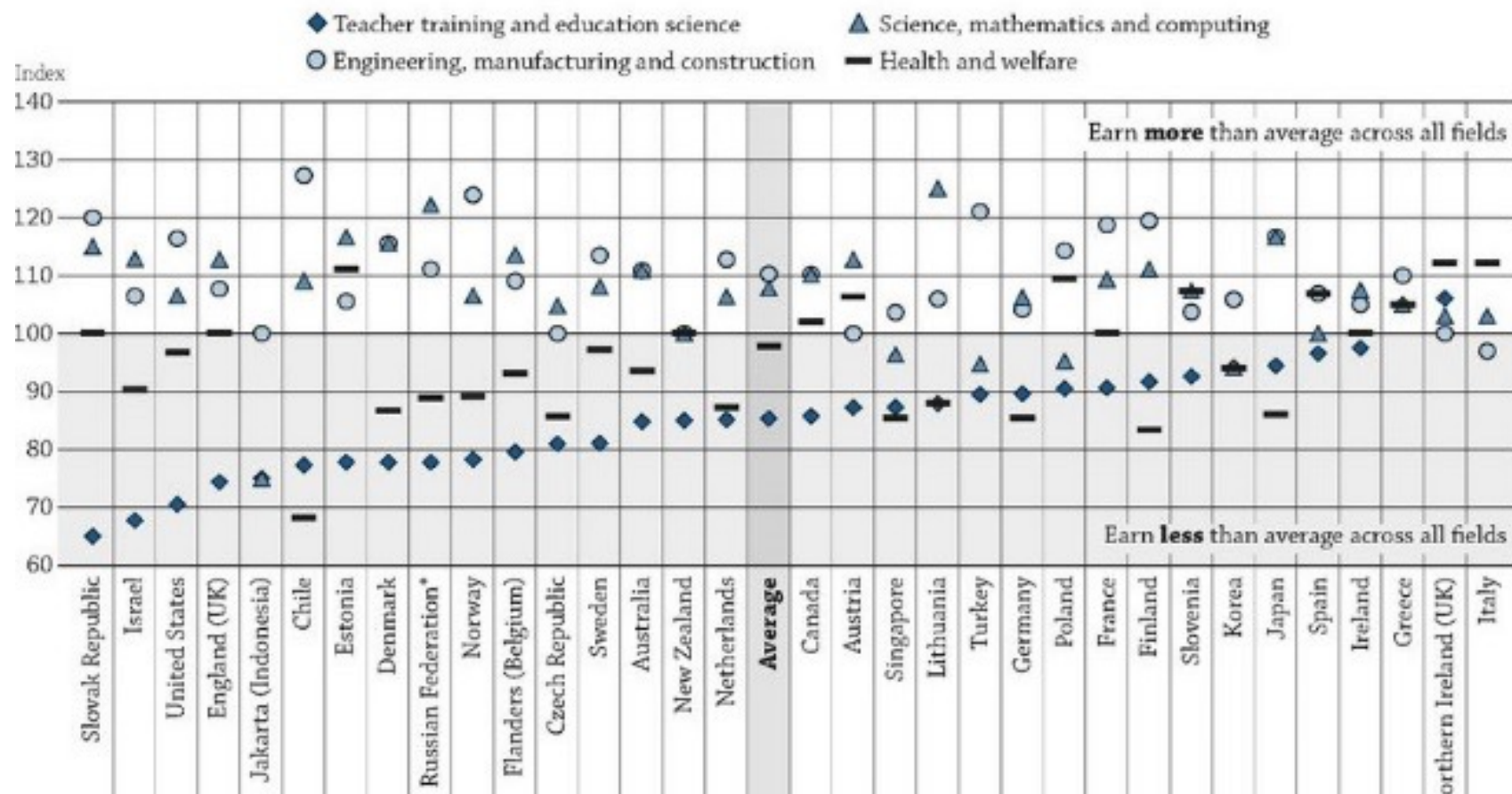
- Below Upper Secondary Education:
 - Adults with below upper secondary education usually have the lowest earnings compared to those with higher educational attainment.
 - They may also face a higher risk of unemployment or underemployment, which can further affect their earnings potential.
- Upper Secondary Education:
 - Individuals with upper secondary education generally earn more than those with less than upper secondary education.
 - The earnings premium for this group, compared to those with below upper secondary education, can be significant, but it's often less than the premium for tertiary education.

Relative Earnings by Educational Attainment:

- Tertiary Education (Bachelor's degree or equivalent):
 - Earnings tend to jump considerably for those with a bachelor's degree or equivalent when compared to those with only upper secondary education.
 - In many countries, the earnings premium for having a bachelor's degree can be substantial, reflecting the higher skills and expertise associated with tertiary education.
- Advanced Degrees (Master's, Doctorate, or equivalent):
 - Individuals with advanced degrees usually earn even more than those with a bachelor's degree, though the exact premium can vary.
 - The earnings differential between bachelor's degree holders and advanced degree holders can depend on fields of study, labor market conditions, and the specific demands of certain professions.

Figure A6.4. Relative earnings of adults with tertiary education, by field of education studied (2012 or 2015)

Survey of Adult Skills, 25-64 year-old non-students full-time workers; all fields of education = 100



Relative Earnings by Fields of Tertiary Education:

- Teacher Training and Education Science:
 - Graduates typically earn around or slightly above the average for tertiary-educated individuals. Earnings vary based on the level of education they teach (primary, secondary, tertiary) and regional pay structures.
- Science, Math, and Computing:
 - Graduates in these fields often command earnings above the average for tertiary-educated individuals. Those in specialized roles or sectors with high demand, like technology or data analysis, can have particularly high earnings.
- Engineering, Manufacturing, and Construction:
 - These graduates typically have some of the highest earnings among tertiary-educated adults. The demand for engineering and specialized manufacturing skills, especially in regions with strong industries or infrastructure projects, drives these higher earnings.
- Health and Welfare:
 - Health professionals, especially doctors and pharmacists, generally have high earnings. However, those in caregiving or welfare roles might earn less relative to other tertiary-educated peers, though still often above the average for all tertiary fields.

II. Types of Education

1. Formal education: structured and organized, often leading to a recognized certification or degree. It follows a specific curriculum and is guided by institutional rules and regulations.
 - Schools
2. Nonformal education: organized and can be short- or long-term, but it doesn't typically lead to formal qualifications. It's more flexible than formal education and caters to specific learning needs or skill sets.
 - Adult education
 - Training

II. Types of Education

3. Informal education: a lifelong process where individuals acquire knowledge, skills, attitudes, and values from daily experiences and exposure to their surroundings.
 - Self study: involve reading books, online courses, or using various resources to acquire knowledge;
 - learn at home or in community;
 - on the job learning: experiential and happens as individuals tackle work tasks, solve problems, or engage with colleagues.

III. Education and Development

- Education produces:
 - A. Durable consumption good
(e.g., ability to enjoy a novel)
 - B. Human Capital
bring returns in the future

III. Education and Development

- Numerous studies have shown that education and economic growth are positively related
 - Skilled Workforce.
 - Innovation and Research.
 - Social Benefits.
 - Attracting Investments.

Figure A6.1. Relative earnings of adults working full time, by educational attainment (2014)

25-64 year-olds with income from employment; upper secondary education = 100

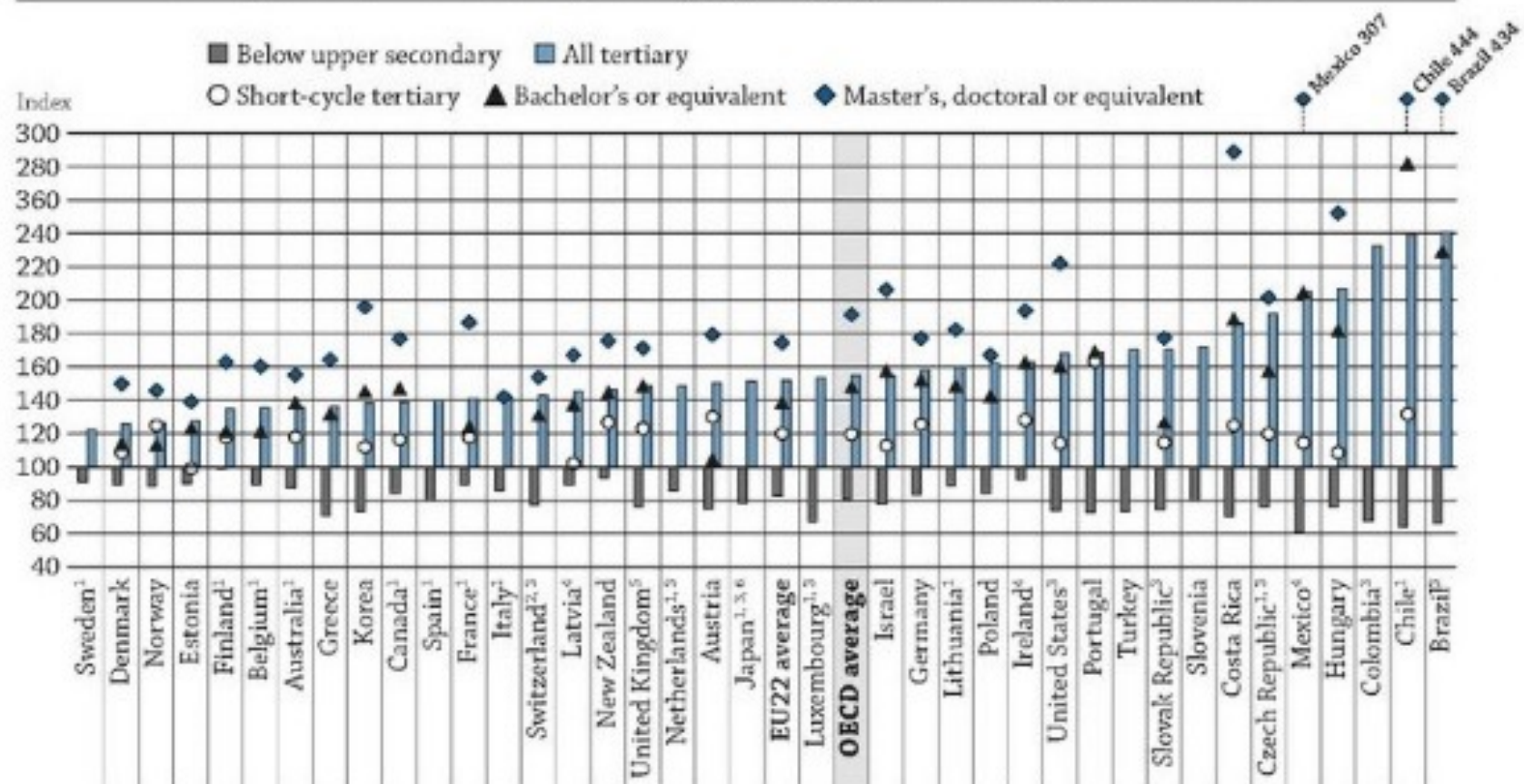
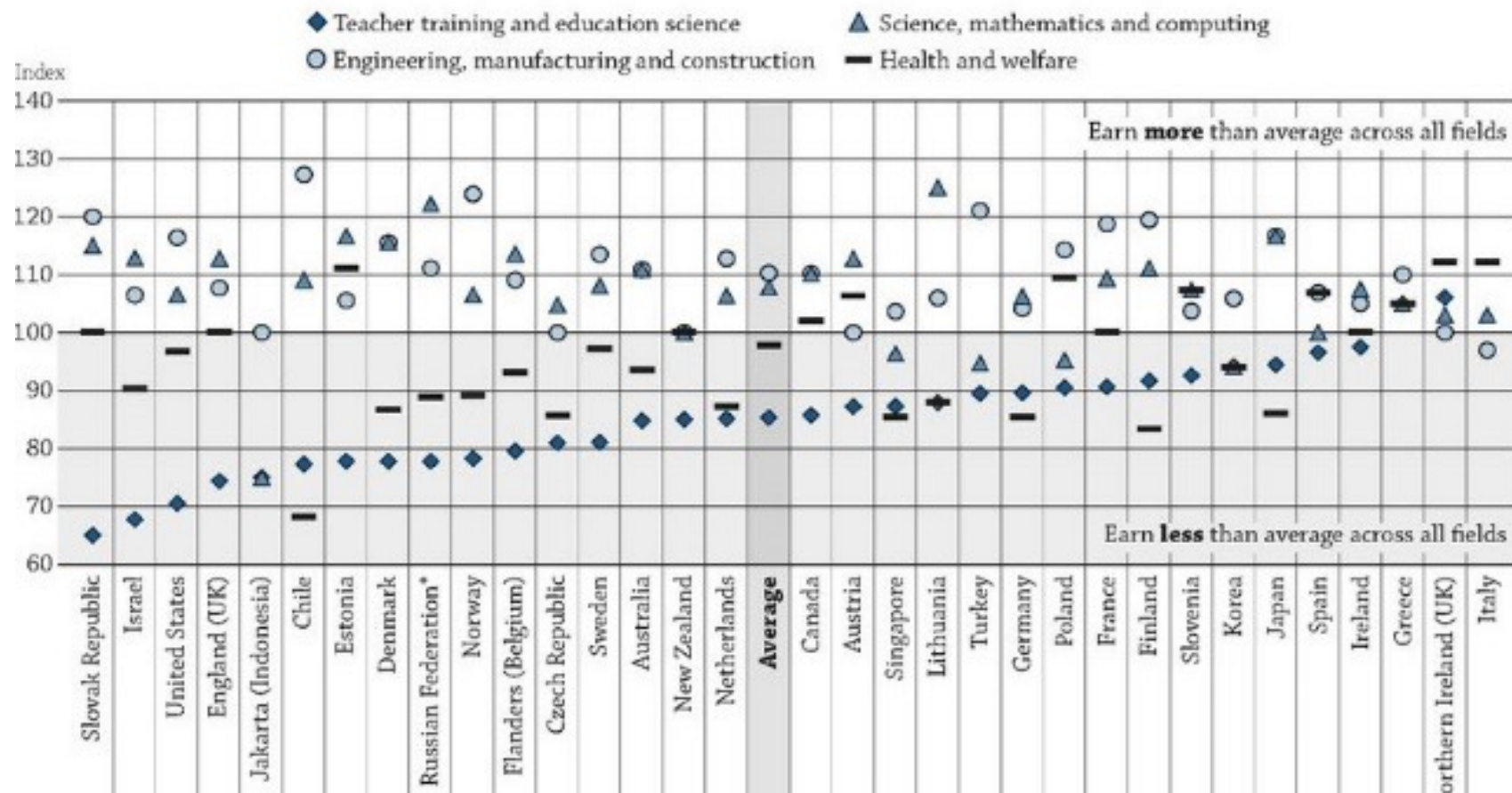


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IV. Invest in Education

1. Cost. The PV of cost in any future year_t:

$$PVC_0^t = \frac{C_t}{(1+i)^t} \quad C_t \text{ includes explicit \& implicit costs.}$$

PVC from year 0 to year n:

$$PVC = \sum_{t=0}^n \frac{C_t}{(1+i)^t} \quad \begin{array}{l} n = 0, 1, 2, \dots \\ i = \text{rate of interest} \end{array}$$

2. Benefit

PV of benefits in any future year_t:

$$PVB_0^t = \frac{E_t}{(1+i)^t} \quad \begin{array}{l} E_t \text{ benefit in } t, \\ t=0,1,2, \dots, n \end{array}$$

PVB from year 0 to year n:

$$PVB = \sum_{t=0}^n \frac{E_t}{(1+i)^t} \quad n = 0,1, \dots$$

PVB

PVC

- If discounted benefits $>$ discounted costs, then investment should be made.

Otherwise, no.

- Internal rate of return: r satisfies.

$$\sum_{t=0}^n \frac{E_t}{(1+r)^t} = \sum_{t=0}^n \frac{C_t}{(1+r)^t}$$

Or

$$\sum_{t=0}^n \frac{E_t - C_t}{(1+r)^t} = 0$$

If $r > i$, invest;

$r < i$, no.

- Example:

	year 0	year 1	PV Total
Returns (Et)	0	110	$\frac{110}{1+r}$
Costs (Ct)	100	0	100

Setting
$$\sum_{t=0}^n \frac{E_t}{(1+r)^t} = \sum_{t=0}^n \frac{C_t}{(1+r)^t}$$

i.e.,
$$\frac{110}{1+r} = 100$$

$$\Rightarrow 110 = 100(1+r)$$

$$\Rightarrow 1+r = 1.1$$

$$\Rightarrow r = 0.1 = 10\%$$

- Example:

	year 0	year 1	year 2	PV Total
Returns (Et)	0	110	121	$\frac{110}{1+r} + \frac{121}{(1+r)^2}$
Costs (Ct)	100	110	0	$100 + \frac{110}{1+r}$

- r: internal rate of returns

Setting $\frac{110}{1+r} + \frac{121}{(1+r)^2} = 100 + \frac{110}{1+r}$

$$\Rightarrow \frac{121}{(1+r)^2} = 100$$

$$\Rightarrow (1+r)^2 = 1.21$$

$$\Rightarrow 1+r = 1.1$$

$$\Rightarrow r = 0.1 = 10\%$$

Social or economic rate of return using :

- all costs, public & private
- all benefits, private & social

Private rate of return versus social rate of return:

- Example:

	Earnings (\$)	
	period 1	period 2
No school	\$15,000	\$25,000
With school	\$0	\$45,000
Gain from schooling	-\$15,000	\$20,000

- The present value of the gain from schooling

$$V = -\$15,000 + \frac{\$20,000}{(1 + i)}$$

- i = discount rate.

if $i = 10\%$, then

$$V = -\$15,000 + \frac{\$20,000}{1 + 0.1} = \$3,181.8$$

- Costs:

Earnings forgone: opportunity Cost \$ 15,000

Explicit Cost: \$3,000 $\left\{ \begin{array}{l} \$1,000 \text{ private} \\ \$2,000 \text{ govt} \end{array} \right.$

The private internal rate of return: The discount rate for which the present value of net benefits equals zero.

Net Benefits = $V - \$1,000$

$$\begin{aligned} &= \frac{\$20,000}{(1+r)} - \$15,000 - \$1,000 \\ &= \frac{\$20,000}{(1+r)} - \$16,000 \end{aligned}$$

Setting Net Benefits = 0

$$\Rightarrow 1+r = \frac{\$20,000}{\$16,000} = 1.25$$

$$\Rightarrow r = 25\%$$

- The Social Rate of Return: The discount rate for which the present value of net social benefits equals zero.

$$\begin{aligned}
 \text{Net Social Benefits} &= V - \$3,000 \\
 &= \frac{\$20,000}{(1+r)} - \$15,000 - \$3,000 \\
 &= \frac{\$20,000}{(1+r)} - \$18,000
 \end{aligned}$$

$$\begin{aligned}
 \text{Net Social Benefits} &= 0 \\
 \Rightarrow 1+r &= \frac{\$20,000}{\$18,000} = 1.111 \\
 \text{Thus, } r &= 11.1\%
 \end{aligned}$$

- The value of human Capital:

$$NPV = \sum_{t=0}^n \frac{E_t}{(1+i)^t}$$

How to make schooling more productive

- Making Schooling Productive: The Hurdles
 - Uneven quality of education
 - Outdated curricula
 - Lack of resources and infrastructure
 - Teacher shortages and unprepared educators

How to make schooling more productive

- Strategies for More Productive Schooling:
 - Curricular Reforms: Modern, relevant, and inclusive syllabi
 - Teacher Training: Continuous professional development
 - Technological Integration: Digital tools and online resources
 - Personalized Learning: Catering to individual student needs

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How to make schooling more productive

- Engaging All Stakeholders:
 - Governments: Policies, funding, and regulations
 - Educators: Implementing best practices and innovations
 - Parents & Communities: Participation and support
 - Private Sector: Investments, partnerships, and technological solutions

How to make schooling more productive

- Successful Interventions:
 - Finland's holistic education model
 - Singapore's focus on teacher quality
 - Tech-driven solutions in South Korea
 - Community schools in rural Africa

Randomized Control Trials and how to reduce the cost of going to school